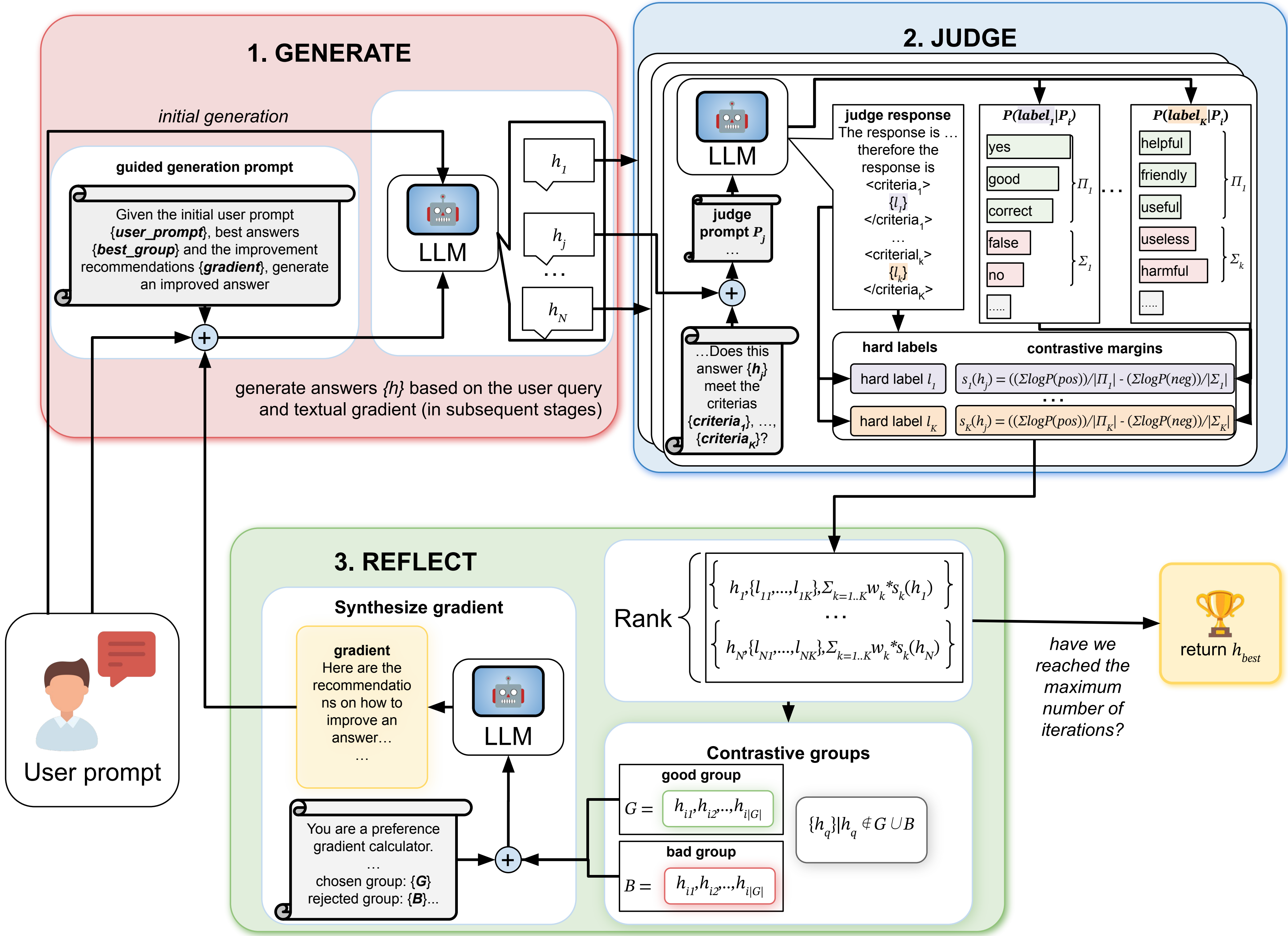

1. GENERATE
initial generation
guided generation prompt
Given the initial user prompt {user_prompt}, best answers {best_group} and the improvement recommendations {gradient}, generate an improved answer
LLM
h_1
h_j
h_N
generate answers {h} based on the user query and textual gradient (in subsequent stages)
2. JUDGE
LLM
judge prompt P_j …
…Does this answer $\{h_j\}$ meet the criterias $\{criteria_1\}$, …, $\{criteria_K\}$?
judge response
The response is … therefore the response is <criteria$_1$> $\{l_1\}$ </criteria$_1$> … <criterial$_k$> $\{l_k\}$ </criteria$_K$>
$P(label_1|P_i)$
yes
good
correct
false
no
.....
Π_1
Σ_1
$P(label_K|P_i)$
helpful
friendly
useful
useless
harmful
.....
Π_1
Σ_k
hard labels
contrastive margins
hard label l_1
$s_1(h_j) = ((\Sigma logP(pos))/|\Pi_1| - (\Sigma logP(neg))/|\Sigma_1|$
hard label l_K
$s_K(h_j) = ((\Sigma logP(pos))/|\Pi_K| - (\Sigma logP(neg))/|\Sigma_K|$
3. REFLECT
Synthesize gradient
gradient
Here are the recommendations on how to improve an answer…
…
LLM
You are a preference gradient calculator.
…
chosen group: {G}
rejected group: {B}...
Rank
$\{h_1, \{l_{11},...,l_{1K}\}, \Sigma_{k=1..K} w_k * s_k(h_1)\}$
…
$\{h_N, \{l_{N1},...,l_{NK}\}, \Sigma_{k=1..K} w_k * s_k(h_N)\}$
Contrastive groups
good group
$G = h_{i1}, h_{i2}, .., h_{i|G|}$
bad group
$B = h_{i1}, h_{i2}, .., h_{i|G|}$
$\{h_q\} | h_q \notin G \cup B$
have we reached the maximum number of iterations?
return h_{best}
User prompt